

Yingshi Huang

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EDUCATION	University of California, Los Angeles (UCLA) , California, United States	
	Ph.D. in Education	Sep. 2023 - Present
	M.S. in Statistics & Data Science	Sep. 2024 - Present
	• Research Interest: Latent variable modeling; Social networks; Casual inference • Advisor: Prof. Minjeong Jeon and Prof. Qing Zhou	
	Beijing Normal University , Beijing, China	
	M.A. in Psychology	Sep. 2020 – Jun. 2023
	South China Normal University , Guangzhou, China	
	B.S. in Psychology	Sep. 2016 – Jun. 2020
PUBLICATIONS	(* correspondent author, + co-first author)	
	Educational and Psychological Measurement	
	1. Zhang, L., Chen, P.*, Huang, Y., Jeon, M. (under review). Uncovering residual structure with variational autoencoders. 2. Huang, Y., Wang, S., Pan, Y., Lu, X., & Chen, P.* (under review). A motivation-based cognitive diagnostic model for disengaged responses detection. 3. Yuan, L., Huang, Y., & Chen, P.* (2025). Online calibration for multidimensional CAT with polytomously scored items: A neural network-based approach. <i>Journal of Educational and Behavioral Statistics</i>, 0(0). doi:10.3102/10769986251315531. 4. Chen, P.*, Dai, Y., Huang, Y. (2023). Test mode effect: Sources, detection, and applications (in Chinese). <i>Advances in Psychological Science</i>, 31(10), 1966–1980. doi: 10.3724/SPJ.1042.2023.01966 5. Yuan, L., Huang, Y., Li, S., & Chen, P.* (2023). Online calibration in multidimensional computerized adaptive testing with polytomously scored items. <i>Journal of Educational Measurement</i>, 60(3), 476-500. doi: 10.1111/jedm.12353. 6. Huang, Y., Ren, H., & Chen, P.* (2023). Item selection methods with exposure and time control for computerized classification test. <i>British Journal of Mathematical and Statistical Psychology</i>, 76, 52–68. doi: 10.1111/bmsp.12281. 7. Ren, H., Huang, Y., & Chen, P.* (2022). Types, characteristics and application of termination rules in computerized classification testing (in Chinese). <i>Advances in Psychological Science</i>, 30(5), 1168. doi: 10.3724/SPJ.1042.2022.01168.	
	Substantive Topics	
	1. Tien, I., Luo, J., Huang, Y., & Jeon, M.* (2026). Visualizing sex differences in school aged youth on the ADOS-2 using a latent space item response model research in Autism. <i>Research in Autism</i>, 136, 202948. doi:10.1016/j.reia.2026.202948. 2. Tien, I.*, Wolpe, S., Huang, Y., Sozeri, S., Lee, M., & Jeon, M. (2025). How diagnostically accurate are #Autism portrayals? A latent space item response modeling approach. <i>Journal of Autism and Developmental Disorders</i>, 1-14. doi:10.1007/s10803-025-07164-5. 3. Huang, Y., Luo, J., Shetty, V., & Jeon, M.* (2025). The power of social talk: A longitudinal network analysis of conversations in fostering interdisciplinary collaboration. <i>Journal of Clinical and Translational Science</i>, 1–26. doi:10.1017/cts.2025.10124.	

4. Ren, J., Luo, J., **Huang, Y.**, Shetty, V., & Jeon, M.* (2025). Mapping the mHealth nexus: A semantic analysis of mHealth scholars' research propensities following an interdisciplinary training institute. *Applied Sciences*, 15(11), 6252. doi: 10.3390/app15116252.
5. Ni, Y.*, **Huang, Y.**, Zhao, S., Ma, Y., & Fang, M. (2025). Development and evaluation of the shortened Chinese version of school social behavior scales-2 using item response theory and psychometric network analysis. *Measurement and Evaluation in Counseling and Development*, 1-19. doi: 10.1080/07481756.2025.2499965.
6. Lu, A.*+, Deng, R.+, **Huang, Y.**+, Song, T., Shen, Y., Fan, Z., & Zhang, J. (2022). The roles of mobile app perceived usefulness and perceived ease of use in app-based Chinese and English learning flow and satisfaction. *Education and Information Technologies*, 1-22. doi: 10.1007/s10639-022-11036-1.
7. Ni, Y., Tein, J. Y., Zhang, M.*, Zhen, F., Huang, F., **Huang, Y.**, Yao, Y., & Mei, J. (2020). The need to belong: A parallel process latent growth curve model of late life negative affect and cognitive function. *Archives of Gerontology and Geriatrics*, 89, 104049. doi: 10.1016/j.archger.2020.104049.

CONFERENCE PRESENTATIONS

1. **Huang, Y.**, & Jeon, M. (2026, April) *ADAM estimation for the hierarchical joint model of response accuracy and response time*. Talk presented at the Annual Meeting of the American Educational Research Association, Los Angeles, CA.
2. **Huang, Y.**, Luo, J., & Jeon, M. (2026, April) *Dimension selection guideline for latent space item response models*. Talk presented at the Annual Meeting of the National Council on Measurement in Education, Los Angeles, CA.
3. **Huang, Y.**, Luo, J., Molenaar, D., & Jeon, M. (2025, April) *Capturing item preknowledge with a response time latent space model framework*. Talk presented at the Annual Meeting of the National Council on Measurement in Education, Denver, CO.
4. **Huang, Y.**, Wang, S., Pan, Y., Lu, X., & Chen, P. (2024, April) *A motivation-based cognitive diagnostic model for insufficient responses detection*. Talk presented at the Annual Meeting of the National Council on Measurement in Education, Philadelphia, PA.
5. Yuan, L., **Huang, Y.**, & Chen, P. (2023, July). *Online calibration for P-MCAT: A neural network based approach*. Talk presented at the International Meeting of the Psychometric Society, College Park, Maryland.
6. **Huang, Y.**, Zhang, T., & Chen, P. (2022, July). *Exploring the structure of speed in cognitive diagnostic models*. Poster presented at the International Meeting of the Psychometric Society, Bologna, Italy.
7. Zhang, T., **Huang, Y.**, & Xin, T. (2022, July). *An analysis of adaptive learning recommendation based on reinforcement learning*. Poster presented at the International Meeting of the Psychometric Society, Bologna, Italy.
8. **Huang, Y.**, Ren, H., & Chen, P. (2022, April). *New item selection designs for computerized classification test*. Flash Talk presented at the Annual Meeting of the National Council on Measurement in Education, San Diego, CA, Virtual Meeting.
9. Ren, H., **Huang, Y.**, & Chen, P. (2021, October). Who has the final say in computerized classification test: three types of termination rules. Talk presented at the 23rd National Academic Conference of Psychology, Inner Mongolia.

FUNDED GRANTS

2021 Independent Project (Grant No. BJZK-2020A2-20011).	PI
<i>Item Selection Methods for Computerized Classification Testing</i>	2021 – 2022
Funded by Collaborative Innovation Center of Assessment for Basic Education Quality	\$1,000
2020 National Natural Science Foundation of China (Grant No. 32071092).	Participant
<i>Online Calibration in Computerized Adaptive Testing: New Challenges and Solutions</i>	2021 – 2024

RESEARCH AND
EMPLOYMENT
EXPERIENCE

- Funded by National Natural Science Foundation of China \$90,000
- 2019 Extracurricular Research Project (Grant No. 19XLGA01).** **PI**
High Quality of Teacher-Student Relationships Promotes Mathematics Achievement of Junior High School Students: The Multiple Mediation Effect
 2019 – 2020
 Funded by South China Normal University \$500
- 2018 Special Funds for Seed Breeding Program (Grant No. 18XLGA07).** **Co-PI**
Second Language Learning through Mobile Devices: The Effect of Perceived Usefulness, Perceived Ease of Use, and Flow Experience
 2018 – 2019
 Funded by South China Normal University \$500
- American Board of Internal Medicine (ABIM), Philadelphia, United States**
Psychometric Intern Jun. 2026 – July. 2026
 ABIM aims to enhance the quality of health care through the development and administration of assessments for the certification of internists and subspecialists
- Extended the hierarchical framework to jointly model accuracy, response time, and confidence ratings, leveraging rich multi-source data at ABIM
 - Developed a unified likelihood ratio test for item preknowledge detection to address challenging operational settings
 - Developed two R packages implementing the model estimation and the proposed test statistic
 - Presented the project to the leadership team and broader department
- Jeon lab, Los Angeles, United States**
Graduate Student Researcher Dec. 2023 – Present
 Prof. Minjeong Jeon's lab centers on developing, applying, and estimating various latent variable models for studying measurement and growth
- Developing cultural relevant reading assessment for visually impaired children
 - Analyzing the NIH-supported annual mHealth Training project with TERGMs
 - Developing longitudinal latent process model with multiple learning targets
- National Assessment Center for Key Technologies, Beijing, China**
Research Assistant Sep. 2020 – Jun. 2023
 Prof. Ping Chen's lab focuses on building and developing more effective and efficient algorithms for parameter estimation and adaptive testing
- Designed and conducted studies introducing new item selection methods, online calibration designs, and a motivation-based cognitive diagnosis model
 - Helped manage the pretesting of items for the National Assessment
- National Assessment Guangdong Sub-Center for Item Bank Construction, Guangzhou, China**
Research Assistant Sep. 2017 – Jun. 2020
 Prof. Minqiang Zhang's lab studies the validity of new Verbal and Mathematics item formats, the score reports system, and test equating in China's College Entrance Examination
- Contributed to the Test Quality Analysis Report (Math) of China's College Entrance Exam
 - Assisted with the implementation of the Guangzhou Sunshine education evaluation project and helped write the survey report
 - Designed a study on how students' mathematics performance is affected by the quality of the teacher-student relationship, analyzed data from 3,997 students, and led a funded grant
- Guangdong Key Laboratory of Mental Health and Cognitive Science, Guangzhou, China**
Data Analysis Research Assistant Oct. 2018 – May. 2019
 Prof. Aitao Lu's lab studies the mechanism of language acquisition and how human brains execute language processing
- Designed and conducted studies investigating app-based second-language learning, especially

- for Chinese and English learners
- Collected 786 questionnaires and analyzed the data with latent profile analysis, hierarchical regression, structural equation modeling, and network analysis

TEACHING EXPERIENCE

UCLA, California, United States. *Teaching Assistant*
 EDUC M231E: Statistical Analysis with Latent Variables Jan. 2026 – Mar. 2026
 EDUC 230C: Linear Statistical Models: Analysis of Designed Experiments Mar. 2025 – Jun. 2025
 EDUC 230B: Linear Statistical Models: Multiple Regression Analysis Jan. 2025 – Mar. 2025
 EDUC 230A: Introduction to Research Design and Statistics Sep. 2024 – Dec. 2024
 EDUC 151: Quantitative Research in Education Sep. 2024 – Dec. 2024; Sep. 2025 – Dec. 2025

Beijing Normal University, Beijing, China. *Teaching Assistant*
 Adaptive testing and diagnostic adaptive assessment (graduate course) Sep. 2021 – Jan. 2022
 Think and act like a psychometrician (undergraduate course) Feb. 2021 – Jun. 2021

SKILLS

- Analytical:** Proficient in R, *Mplus*, SPSS; Capable of Python
- Design:** Photoshop, HTML, $\text{\LaTeX}$, Unity

HONORS AND AWARDS

Graduate Summer Research Mentorship (GSRM) Award, UCLA 2026
 Graduate Dean's Scholar Award, UCLA 2024, 2025
 Gordon & Olga Smith Scholarship, UCLA 2023
 Outstanding Graduates of Beijing, Beijing Municipal Education Commission 2023
 Outstanding Graduates, Beijing Normal University 2023
 National Scholarship, Minister of Education of China 2022
 First-class Academic Scholarship, Beijing Normal University 2021
 Second-class Comprehensive Scholarship, South China Normal University 2017 - 2018

ADMINISTRATIVE AND OTHER EXPERIENCE

Editorial Assistant, Journal of Educational and Behavioral Statistics (JEBS) Jan. 2026 – Present

Reviewer Experience:
 Current Psychology; Journal of Educational Measurement (JEM); National Council on Measurement in Education (NCME) Annual Meeting 2025; American Educational Research Association (AERA) Annual Meeting 2024, 2025, 2026

Vice Chairman, The Business Pioneering Camp 2018 – 2019
 Planned and organized business competitions, trained new members, and managed the budget (Awarded as Outstanding Student Leader)

Volunteer Experience:
 Devoted to supporting education at the Wanzi Primary School in Meizhou, an undeveloped county in Southeast China (Awarded as Advanced Individual) [\[Program Web\]](#) Aug. 2017